Case-disjoint benchmark of surrogate models for flow and thermal fields in supercritical CO_2 natural-circulation loops

Abstract

Surrogate models for supercritical carbon dioxide (sCO_2) natural-circulation loops (NCLs) can replace day-long steady-state conjugate computational fluid dynamics (CFD) simulations with fast post-training field evaluations, yet most published evaluations rely on random node-level train–test splits or single pooled metrics that mask generalization failure across unseen operating conditions. Here we present a case-disjoint surrogate benchmark built from 42 legacy steady-state RANS simulations solved with a k–ε family turbulence model (k and ε are exported) and a Span–Wagner equation of state verified for internal thermodynamic consistency against `CoolProp` (median 0.0008%) across three heater–cooler configurations (`HHHC`, `HHVC`, and `VHVC`) and two loop scales. Across three random seeds, we benchmark 19 surrogate model variants (16 cross-configuration pooled models, including multiplicative `Domain-informed FiLM-MLP` and its trunk-matched input-concatenation ablation `Domain-informed Concat-MLP`, plus 3 within-group mesh-aligned models: `POD-RBF`, `POD-GPR`, and `Nearest case`) alongside a specified $64 \times 64 \times 16$ voxel round-trip reconstruction reference (`Lattice floor (oracle)`) under five-fold stratified case-disjoint cross-validation and leave-one-group-out structural/regime shift. We show that random node splitting understates five-fold case-disjoint error across the three strictly matched-configuration tree baselines (`Random Forest`, `Extra Trees`, `CatBoost`) by $6.78\times$–$8.25\times$ on density, $11.29\times$–$30.08\times$ on gauge pressure, and $5.62\times$–$10.24\times$ on temperature in case-macro MAE ($6.78\times$–$10.37\times$ and $11.29\times$–$87.54\times$ when including `XGBoost`) because high-capacity models memorize case-specific operating levels from adjacent mesh nodes. We also show that separating closed-loop gauge pressure into its case-level thermal-pressurization offset and centered spatial residual—benchmarked against a zero-residual baseline—prevents small within-case hydrostatic variations from distorting model rankings. Across the eight exported CFD fields, no single architecture wins universally: among cross-configuration pooled models, `Domain-informed Concat-MLP` achieves the lowest case-macro MAE on density (26.3 kg m^{-3}) and temperature (5.14°C) whereas `DeepONet` (0.837) and `PINN continuity` (1.350) achieve the lowest median case nMAE on those fields; `Extra Trees` best predicts case-level gauge pressure (83.2 kPa case-macro MAE); and multiplicative feature-modulated coordinates (`Domain-informed FiLM-MLP`) achieve the lowest case-macro MAE and median case nMAE among pooled models across both turbulence fields (k, ε) and all three velocity components (u, v, w), while within-group mesh-aligned `POD-RBF` and `POD-GPR` achieve the lowest median case nMAE on temperature (0.974), ε (0.150 for `POD-GPR`; 0.177 for `POD-RBF`), and k (0.237 for `POD-RBF`; 0.245 for `POD-GPR`) when interpolating on a fixed loop mesh.

Keywords: supercritical CO_2, natural-circulation loop, CFD surrogate benchmark, case-disjoint validation, neural operators, proper orthogonal decomposition

1 Introduction

Natural-circulation loops (NCLs) operating with supercritical fluids remove heat without mechanical pumps by balancing buoyancy forces between heated and cooled sections against frictional and local losses [1, 2, 3]. Supercritical carbon dioxide (sCO_2, critical point $T_c = 304.13$ K $= 30.98$°C, $p_c = 7.3773$ MPa $= 73.773$ bar) is widely used as a scaled simulant fluid for supercritical water reactor (SCWR) thermal-hydraulics and compact power cycles because its lower critical point permits experiments and conjugate CFD studies near the pseudocritical locus [4, 5, 6]. Near the pseudocritical temperature, steep variations in density ρ, specific heat c_p, viscosity μ, and thermal conductivity k strongly couple buoyancy gradients with turbulence production and heat transfer [7, 8, 9].

Because resolving conjugate heat transfer between the primary sCO_2 pipe, the solid metallic wall, and the secondary cooling-water jacket requires fine near-wall meshes and real-gas property evaluations, each steady-state RANS simulation in our dataset requires roughly 24 hours of compute time. Data-driven spatial surrogates—including gradient-boosted trees [10, 11, 12], coordinate neural networks [13, 14, 15], neural operators [16, 17, 18, 19], and proper orthogonal decomposition (POD) reduced-order models [20, 21]—provide fast post-training field evaluations once fit. However, three practical issues complicate surrogate benchmarking on closed natural-circulation loops: (i) random node-level train–test splits leak case-specific operating levels across neighboring mesh nodes; (ii) pooled metrics such as R^2 conflate between-case operating-level shifts with within-case spatial profile accuracy, while naive within-case normalization breaks down for gauge pressure in closed loops; and (iii) slender circular pipes ($D_i/L \sim 10^{-2}$) and varying heater–cooler orientations create different inductive-bias requirements for thermodynamic state fields versus velocity and turbulence fields.

To address these issues, this paper contributes: (1) a controlled comparison of 80/20 random-node splitting against five-fold case-disjoint cross-validation across three strictly matched-configuration tree ensembles, `XGBoost`, and two linear/constant controls; (2) a physical characterization and two-part evaluation protocol for closed-loop gauge pressure referenced against a zero-residual baseline; and (3) a three-seed benchmark of 19 surrogate model variants (16 cross-configuration pooled models and 3 within-group mesh-aligned models, plus a specified $64 \times 64 \times 16$ voxel round-trip reconstruction reference) across eight CFD fields under both five-fold case-disjoint interpolation and leave-one-group-out structural/regime shift.

2 Physical Domain and sCO_2 Natural-Circulation Loop Dataset

2.1 Loop Geometry, CFD Solver Configuration, and Primary Fluid Domain

The dataset consists of 42 legacy converged steady-state RANS conjugate heat-transfer simulations of closed rectangular sCO_2 natural-circulation loops solved in ANSYS Fluent and exported on cross-sectional planes via ANSYS CFD-Post. All simulations use a k–ε family turbulence model (k and ε are exported as RANS closure variables rather than DNS/LES or experimental measurements) and a Span–Wagner equation of state verified for internal thermodynamic consistency against `CoolProp` [4, 22] (median relative density discrepancy 0.0008% between exported (T, p_{abs}) and ρ) for the primary sCO_2 fluid. Because the dataset is curated from archived CFD-Post plane exports, this `CoolProp` audit verifies equation-of-state consistency of the exported fields rather than mesh-independence or experimental flow validation.

The 42 cases cover three heater–cooler configurations—horizontal heater/horizontal cooler (`HHHC`), horizontal heater/vertical cooler (`HHVC`), and vertical heater/vertical cooler (`VHVC`)—across two loop scales (Table 1): 29 small-loop cases ($L_x \times L_y = 0.8936 \times 1.0000$ m, $D_i = 15.00$ mm) and 13 large-loop `VHVC` cases ($L_x \times L_y = 3.0100 \times 4.0800$ m, $D_i = 13.84$ mm). Complete radial dimensions of the stainless-steel wall and cooling-water jacket, axial heater/cooler spans, and sampling-plane counts (97 planes in small loops; 142 planes in the large loop) are given in Supplementary Section S1.

Each simulation is indexed by four boundary parameters encoded in its case name `{CONFIG}_P{Q}_{p}_bar_{mdot}kgs_T_{Tin}`: heater power $Q \in [10, 2500]$ W, nominal operating pressure $p_{\text{op}} \in \{80, 85, 90, 100\}$ bar, cooling-water mass flow rate $\dot{m} \in [0.010, 0.150]$ kg s^{-1}, and cooling-water inlet temperature $T_{\text{in}} \in \{283.5, 300, 305, 315\}$ K (Fig. 1). At each node (x, y, z), eight CFD target fields are exported: density ρ (kg m^{-3}), gauge pressure p_{gauge} (Pa), temperature T (°C), turbulence eddy dissipation ε (m^2 s^{-3}), turbulence kinetic energy k (m^2 s^{-2}), and velocity components (u, v, w) (m s^{-1}). Removing solid wall nodes (`null` velocity/turbulence, $\rho \approx 7970$ kg m^{-3}) and outer cooling-water jacket nodes ($\rho > 950$ kg m^{-3}) and deduplicating intersecting plane coordinates yields 5,451,341 unique primary sCO_2 nodes ($\rho \in [153.2, 892.5]$ kg m^{-3}; Fig. 2).

Table 1: Summary of the 42 steady-state CFD cases by heater–cooler configuration and loop scale. Full radial and plane-sampling details are in Supplementary Section S1.

Config.	Scale	Cases	$L_x \times L_y$ (m)	D_i (mm)	Q (W)	p_{op} (bar)	$\dot{m}$ (kg s^{-1})	sCO$_2$ nodes/case
HHHC	Small	20	0.894×1.000	15.00	1000–2000	80–100	0.050–0.150	97,264
HHVC	Small	5	0.894×1.000	15.00	10–50	90	0.050	89,771
VHVC	Small	4	0.894×1.000	15.00	10–40	90	0.050	89,647
VHVC	Large	13	3.010×4.080	13.84	250–2500	85	0.010–0.035	207,586
Total	**4 groups**	**42**	–	–	**10–2500**	**80–100**	**0.010–0.150**	**5,451,341 total**

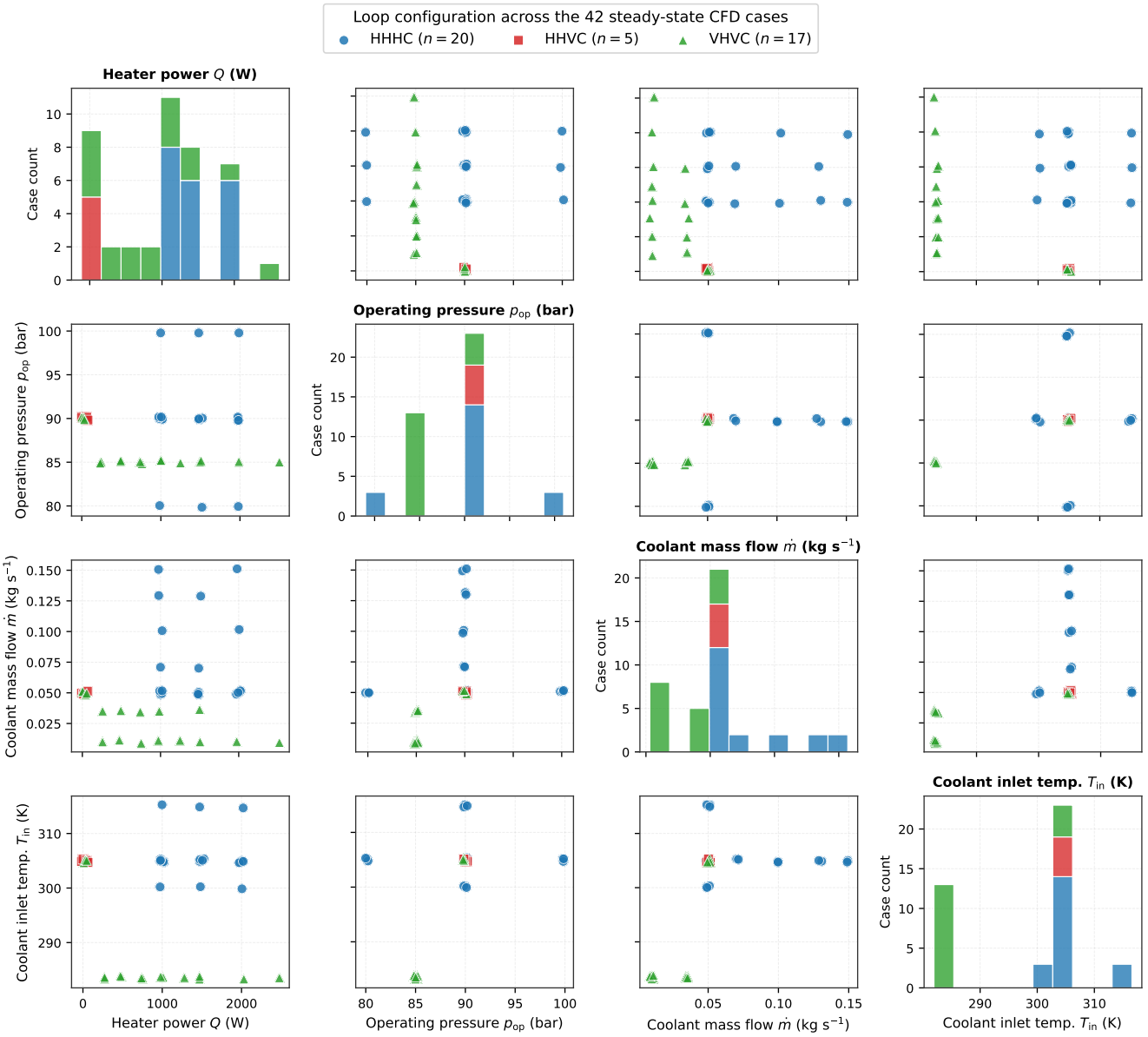


Figure 1: Operating-condition distribution across the 42 steady-state CFD cases by heater–cooler configuration (HHHC, HHVC, VHVC). Diagonal panels show stacked case-count histograms for Q, p_{op}, $\dot{m}$, and T_{in}.

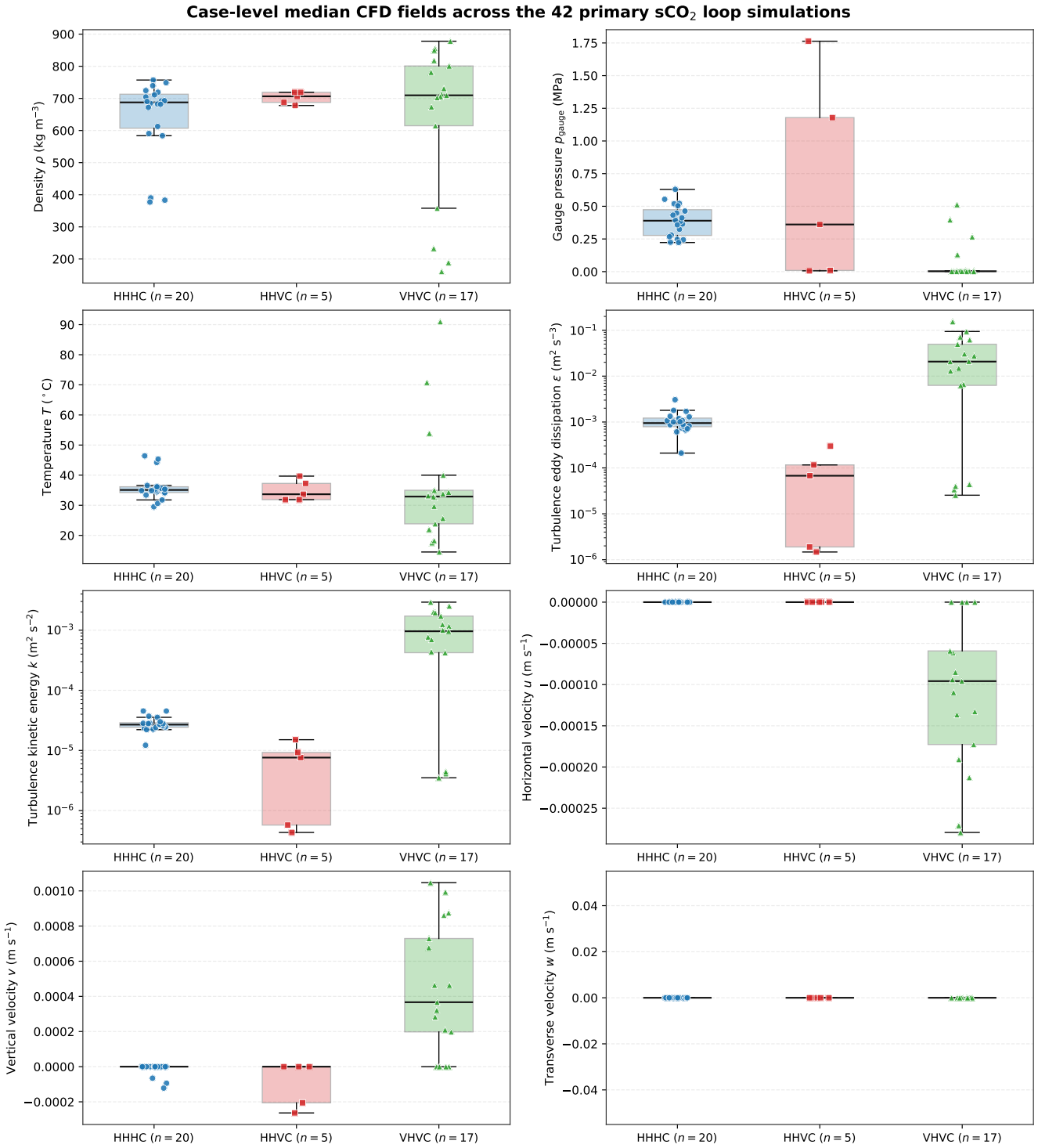


Figure 2: Case-level median distributions of the eight exported CFD fields across the 42 primary sCO_2 loop simulations (5, 451, 341 nodes), grouped by configuration (`HHHC`, `HHVC`, `VHVC`).

2.2 Behavior of Exported Gauge Pressure Across Cases

To separate directly verified properties of the exported CFD files from inferred solver-configuration explanations, we first report the quantities computed directly from the 42 plane-export datasets. In all 42 simulations, the local absolute pressure matching the exported CO_2 density is the sum of the nominal

operating pressure p_{op} and the exported gauge pressure $p_{\text{gauge}}(\mathbf{x})$:

$$p_{\text{abs}}(\mathbf{x}) = p_{\text{op}} + p_{\text{gauge}}(\mathbf{x}). \tag{1}$$

Evaluating $(T(\mathbf{x}), p_{\text{abs}}(\mathbf{x}))$ in `CoolProp` [4, 22] reproduces the exported CFD density $\rho(\mathbf{x})$ across all 42 cases to a median relative difference of 0.0008% (0.00004%–0.0015%). Directly computing the within-case population standard deviation $\sigma_c(p_{\text{gauge}})$ and case mean $\bar{p}_c$ shows a stark scale separation: within any single case c, $\sigma_c(p_{\text{gauge}})$ is only 0.15–0.39 kPa in the 29 small loops and 1.45–8.79 kPa in the 13 large loops (42-case arithmetic mean $\bar{\sigma}_c = 1.51$ kPa; root-mean-square 2.89 kPa), whereas across cases $\bar{p}_c$ spans 1.42 kPa to 1.763 MPa (Fig. 3). Physically, because a full hydrostatic head $\rho g H$ over $H = 1$–4 m would produce 7–30 kPa of vertical variation, we infer that the Fluent setup subtracted a spatially varying hydrostatic reference field $\rho_0 \mathbf{g} \cdot \mathbf{x}$ (whereas subtracting any spatially constant offset would not alter σ_c):

- **Large loop vs. small loops (observed $\bar{p}_c$ ranges and inferred reference convention):** Directly from the exports, the 13 large-loop `VHVC` cases have $\bar{p}_c \in [1.42, 8.77]$ kPa with $p_{\text{gauge}} \approx 0$ Pa near the top of the loop ($y = 4.08$ m; Fig. 3a), whereas in the 29 small-loop cases (`HHHC`, `HHVC`, and small `VHVC`) $\bar{p}_c$ spans 6.1 kPa to 1.763 MPa and rises strongly with the loop-mean temperature $\bar{T}_c$. We infer that the large-loop simulations applied a zero-gauge reference pressure pin near the top boundary, whereas the small-loop simulations evolved without a fixed zero-gauge pin so that $\bar{p}_c$ reflects thermal pressurization relative to the initial state.

- **Observed pressure–temperature slope vs. `CoolProp` constant-density derivative:** Directly fitting $\bar{p}_c$ versus $\bar{T}_c$ across the 17 small-loop cases at $p_{\text{op}} = 90$ bar and $T_{\text{in}} = 305$ K gives an *empirical slope* of 0.214 MPa K^{-1} (0.235 MPa K^{-1} across the five `HHVC` cases; Fig. 3b). By comparison, the theoretical isochoric derivative $(\partial p/\partial T)_\rho$ evaluated in `CoolProp` at each case's exported plane-weighted mean state $(\bar{\rho}_{\text{plane},c}, \bar{T}_c)$ is 0.315–0.365 MPa K^{-1} (median 0.338 MPa K^{-1}; 0.133–0.417 MPa K^{-1} across all 29 small-loop cases in Supplementary Table S0b). Directly inspecting the plane-weighted mean CO_2 density $\bar{\rho}_{\text{plane},c}$ (note that $\bar{\rho}_{\text{plane},c}$ is the arithmetic mean over exported CFD-Post sampling-plane nodes rather than a volume-integrated total mass inventory) shows that $\bar{\rho}_{\text{plane},c}$ varies across cases at the same $(p_{\text{op}}, T_{\text{in}})$: at 90 bar, 305 K it spans 671.6–704.0 kg m^{-3} (SD = 9.79 kg m^{-3}) in `HHHC`, 681.2–717.8 kg m^{-3} (SD = 16.66 kg m^{-3}) in `HHVC`, and 696.8–712.1 kg m^{-3} (SD = 6.62 kg m^{-3}) in small `VHVC` (and 161.5–860.8 kg m^{-3}, SD = 270.83 kg m^{-3} in `VHVC/large`). Because warmer small-loop cases exhibit slightly lower $\bar{\rho}_{\text{plane},c}$ ($\mathrm{d}\bar{\rho}_{\text{plane}}/\mathrm{d}\bar{T} < 0$), first-order expansion $\mathrm{d}\bar{p} \approx (\partial p/\partial T)_\rho \, \mathrm{d}\bar{T} + (\partial p/\partial \rho)_T \, \mathrm{d}\bar{\rho}$ accounts for why the observed across-case slope (0.214 MPa K^{-1}) is lower than the pure constant-density derivative (0.315–0.365 MPa K^{-1}).

- **Near-stagnant steady solutions in `HHVC` at 20 W and 30 W:** Directly from the exported velocity and temperature fields in the five `HHVC` cases at 90 bar and 305 K, the $Q = 20$ W and 30 W runs (`HHVC_P20_90_bar_0.05kgs_T_305` and `HHVC_P30_90_bar_0.05kgs_T_305`) are near-stagnant steady solutions (mean speed $\overline{|\mathbf{u}|} = 0.9$–$1.1$ mm s^{-1}, $\bar{T}_c = 31.92$–31.94°C, $\bar{p}_c = 6.1$–9.2 kPa), whereas the 10 W, 40 W, and 50 W runs have sustained buoyant circulation ($\overline{|\mathbf{u}|} = 0.044$–$0.107$ m s^{-1}, $\bar{T}_c = 33.58$–39.36°C, $\bar{p}_c = 0.361$–1.763 MPa; Fig. 3b).

3 Benchmark Methodology, Models, and Evaluation Protocol

3.1 Validation Protocols, Leakage Prevention, and Evaluation Metrics

Every model predicts the eight target fields $\mathbf{y}(\mathbf{x}, \mathbf{c}) \in \mathbb{R}^8$ at coordinates $\mathbf{x} = (x, y, z)$ given the case descriptor $\mathbf{c} = (Q, p_{\text{op}}, \dot{m}, T_{\text{in}}, L_x, L_y, D_i, \text{one-hot config})$. We evaluate three validation protocols:

1. **Five-fold stratified case-disjoint cross-validation:** For each seed $s \in \{2026, 2027, 2028\}$, the 42 cases are partitioned into five folds (9, 9, 8, 8, 8 test cases) stratified by the four groups (`HHHC/small`,

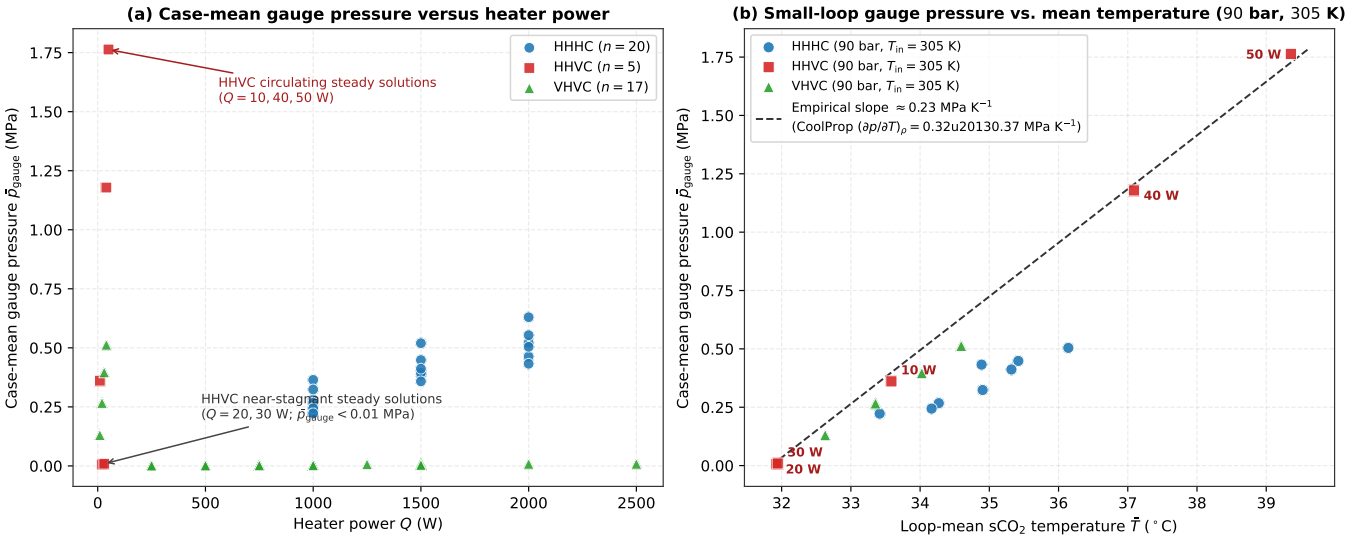


Figure 3: Case-level gauge-pressure behavior across the 42 sCO_2 simulations. (a) Case-mean gauge pressure $\bar{p}_{\text{gauge}}$ versus heater power Q. (b) In the 17 small-loop cases at $p_{\text{op}} = 90$ bar and $T_{\text{in}} = 305$ K, $\bar{p}_{\text{gauge}}$ follows an empirical slope of 0.214 MPa K^{-1} (0.235 MPa K^{-1} across `HHVC`) versus loop-mean temperature $\bar{T}_c$, compared against the `CoolProp` constant-density derivative $(\partial p/\partial T)_\rho = 0.315$–$0.365$ MPa K^{-1} (median 0.338 MPa K^{-1}) evaluated at $(\bar{\rho}_{\text{plane},c}, \bar{T}_c)$.

`HHVC/small`, `VHVC/small`, `VHVC/large`). Each seed independently re-randomizes the five-fold case partition, the four validation cases per fold, the training node subsample (30,000 nodes/case in GPU 3-seed CV; 20,000 in the single-seed CPU split test), and model initialization. Held-out test cases are evaluated on 100% of their mesh nodes (5,451,341 out-of-fold predictions per seed).

2. **Within-group mesh-aligned cross-validation:** Mesh-aligned models (`POD-RBF`, `POD-GPR`, `Nearest case`) require a shared mesh ordering and therefore train and predict strictly within the matching configuration/scale group of each five-fold split.

3. **Leave-one-group-out structural/regime shift:** All cases in one structural group (`HHHC/small`, $n = 20$; `HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$; or `VHVC/large`, $n = 13$) are held out while the 16 cross-configuration pooled models train on the remaining three groups (16 models × 8 fields = 128 model–field evaluations per group). Because each group pairs a distinct heater–cooler arrangement or loop scale with its own operating-parameter range (Table 1), this protocol measures combined structural and operating-regime shift rather than pure geometric extrapolation at fixed operating conditions.

Verified train-only preprocessing and split hygiene: In the benchmark pipeline (`kaggle_clean_case_cv_predictions.ipynb`), four programmatic safeguards prevent data leakage across splits: (i) explicit set-disjointness assertions verify that training, validation, and test case ID sets are pairwise disjoint on every fold and group holdout before any preprocessing (`results/split_assignments.csv`); (ii) validation-case selection (`pick_val`) draws the four validation cases per fold exclusively from strata with at least eight non-test cases (`HHHC/small` and `VHVC/large`), ensuring that the small groups (`HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$) retain all non-test cases (4 and 3–4 training cases per fold) for model fitting; (iii) input feature standardization (`cf_mean`, `cf_scale`, with a zero-variance fallback `cf_scale = 1.0`) and target scaling (`TargetScaler`: $\operatorname{arcsinh}(y/\operatorname{median}(|y_{\text{train}}|))$ followed by z-score standardization on heavy-tailed ε and k, and z-score standardization on the other six fields, inverted via `np.clip(z, -12.0, 12.0)` during prediction) are fit strictly on the fold's training cases (`split["train_ids"]`); and (iv) for within-group mesh-aligned models (`run_group_method`), operating-parameter standardization, snapshot mean subtraction, truncated SVD basis construction (≤ 12 modes, 99.99% energy), and leave-one-out RBF

hyperparameter selection (`tune_rbf`) are computed exclusively on the training cases belonging to that group ($\{c \in \mathcal{I}_{\text{train}} : g(c) = g\}$).

For each held-out case c (N_c nodes, truth $y_{c,i}$, prediction $\hat{y}_{c,i}$, case mean $\bar{y}_c = N_c^{-1}\sum_{i=1}^{N_c} y_{c,i}$, and population within-case standard deviation $\sigma_c = \sqrt{N_c^{-1}\sum_{i=1}^{N_c}(y_{c,i} - \bar{y}_c)^2}$ with `ddof=0`), we compute case-level MAE, RMSE, case-normalized MAE ($\text{nMAE}_c = \text{MAE}_c/\sigma_c$), and node-pooled R^2_{pooled}:

$$\text{MAE}_c = \frac{1}{N_c}\sum_{i=1}^{N_c}|\hat{y}_{c,i} - y_{c,i}|, \quad \text{RMSE}_c = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}(\hat{y}_{c,i} - y_{c,i})^2}, \quad R^2_{\text{pooled}} = 1 - \frac{\sum_{c,i}(\hat{y}_{c,i} - y_{c,i})^2}{\sum_{c,i}(y_{c,i} - \bar{y}_{\text{all}})^2}. \tag{2}$$

For each seed s, we aggregate across the 42 cases via the 42-case arithmetic mean for Macro MAE and Macro RMSE, the 42-case median and 90th percentile (P90) for nMAE_c, and pooled R^2_{pooled} across all $5,451,341$ nodes, and then report the mean ± SD across the three seeds (in Supplementary Tables S1–S7, 95% case-bootstrap confidence intervals are formed by drawing $B = 400$ resamples of the 42 held-out cases with replacement per seed and averaging the 2.5th and 97.5th percentile endpoints across the three seeds). Because $\bar{\sigma}_c(p_{\text{gauge}}) = 1.51$ kPa is small relative to between-case pressure shifts (Section 2.2), we suppress raw nMAE_c for p_{gauge} and decompose each case's pressure error into its case-mean level error $e_{\text{level},c}$ and centered spatial residual $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$ (satisfying $\text{RMSE}_c^2 = (\hat{\bar{p}}_c - \bar{p}_c)^2 + \text{RMSE}_{\text{res},c}^2$ identically under `ddof=0`):

$$e_{\text{level},c} = |\hat{\bar{p}}_c - \bar{p}_c|, \quad \text{RMSE}_{\text{res},c} = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}\left(\tilde{\hat{p}}_{c,i} - \tilde{p}_{c,i}\right)^2}, \quad \text{nRMSE}_{\text{res},c} = \frac{\text{RMSE}_{\text{res},c}}{\max(\sigma_c, 1\text{ kPa})}. \tag{3}$$

For each seed, $e_{\text{level},c}$ is averaged across the 42 cases, $\text{nRMSE}_{\text{res},c}$ is summarized by its 42-case median, and physical-unit residual error is summarized by both the 42-case root-mean-square $\text{RMSE}_{\text{res,RMS}} = \sqrt{\frac{1}{42}\sum_{c=1}^{42}\text{RMSE}_{\text{res},c}^2}$ and the 42-case arithmetic mean $\overline{\text{RMSE}}_{\text{res}} = \frac{1}{42}\sum_{c=1}^{42}\text{RMSE}_{\text{res},c}$ before averaging across the three seeds (both aggregations are reported in Supplementary Table S8). Predicting a spatially constant pressure within each case ($\tilde{\hat{p}}_{c,i} = 0$, achieved by `Mean baseline`, where $\text{RMSE}_{\text{res},c} = \sigma_c$) defines the **zero-residual baseline**, which yields a three-seed mean $\text{RMSE}_{\text{res,RMS}}$ of 2.89 kPa (2886 Pa), a 42-case arithmetic-mean $\overline{\text{RMSE}}_{\text{res}}$ of 1.51 kPa (1513 Pa), and a median $\text{nRMSE}_{\text{res}}$ of 0.202.

3.2 Benchmark Model Suite, Optimization Protocol, and Verified Hyperparameters

Table 2 summarizes all 19 evaluated surrogate model variants (16 cross-configuration pooled models and 3 within-group aligned models) plus the non-competing `Lattice floor (oracle)` voxel round-trip reconstruction reference, together with parameter counts, per-fold execution times, and verified hyperparameters from the GPU and CPU benchmark scripts. All pointwise neural networks train with case-balanced mini-batches of $8,192$ nodes per step (drawing an equal number of nodes from each training case in the fold), cosine learning-rate decay, gradient clipping at 1.0, and early stopping (patience 10, evaluated every 200 steps on the four held-out validation cases per fold, with the inner gradient-descent training loop capped at 2600 steps or 480 s [8 min]/fold on an NVIDIA P100 GPU; the total per-fold wall-clock times reported in Table 2, such as 666 s for `Latent Neural Operator` and 746 s for `GAMS-GNN-style`, additionally include multi-scale kNN/voxel graph construction, validation-set evaluations, and full-mesh out-of-fold inference across 1.1–1.8 million test nodes per fold). To compare multiplicative feature modulation against input concatenation at matched backbone width (256) and depth (5 residual blocks), `Domain-informed FiLM-MLP` ($1,069,064$ parameters) modulates each hidden block by $\mathbf{h}^{(\ell)} \odot (1 + \boldsymbol{\gamma}^{(\ell)}(\mathbf{c})) + \boldsymbol{\beta}^{(\ell)}(\mathbf{c})$ [15], whereas its trunk-matched ablation `Domain-informed Concat-MLP` ($344,072$ parameters; 3.11× fewer parameters because it omits the five per-block linear FiLM generators) concatenates $\mathbf{c}$ at the input layer. In `PINN continuity`, the steady mass-conservation residual is computed in physical units via automatic differentiation with respect to normalized coordinates $\tilde{x}_a \in [0, 1]$ divided by each case's physical bounding-box span $\Delta x_{c,a}$: $\mathcal{R}_{\text{cont}} = S_{\text{flux}}^{-1}\sum_{a\in\{x,y,z\}}(\Delta x_{c,a})^{-1}\partial(\hat{\rho}\hat{u}_a)/\partial\tilde{x}_a$, where $S_{\text{flux}} = (|\mu_\rho| + \sigma_\rho)\bar{\sigma}_u/\overline{\Delta x}$ normalizes the divergence residual before multiplying by $\lambda_{\text{cont}} = 0.05$.

Table 2: Evaluated surrogate models (16 pooled + 3 within-group aligned + 1 voxel round-trip reconstruction reference), parameter counts, mean GPU five-fold total execution time per fold (inner training loop capped at 480 s plus graph construction, validation, and full-mesh test inference), and verified hyperparameters.

Model	Family (time/fold)	Conditioning representation, parameter count, & verified hyperparameters
`Mean baseline`	Constant (4.5 s)	Predicts training-fold node mean per target field (zero-residual baseline $\tilde{p}_{c,i} = 0$ on p_{gauge}).
`Ridge`	Linear (2.2 s)	ℓ_2 linear regression ($\alpha = 10.0$) on standardized $[\mathbf{x}, \mathbf{c}]$.
`Random Forest`	Bagged (35.5 s)	Multi-output CART on $[\mathbf{x}, \mathbf{c}]$: 48 trees (80 in CPU split test; matched across both splits), max depth 20, max leaf nodes 4096 (2048 CPU), min leaf 5, max features 0.8, bootstrap 0.3.
`Extra Trees`	Random (20.2 s)	Multi-output randomized trees on $[\mathbf{x}, \mathbf{c}]$: 64 trees (96 CPU; matched across splits), depth 20, max leaf 4096.
`XGBoost`	Boosted (29.5 s)	Per-target `hist` trees: 300 trees (220 CPU), depth 8 (depth 6 on random-node CPU), lr 0.08, subsample 0.85, colsample 0.85, $\lambda = 2.0$.
`CatBoost`	Oblivious (19.3 s)	Multi-output symmetric trees (`MultiRMSE`): 350 iterations, depth 8, lr 0.08, $\ell_2 = 3.0$ (matched across splits).
`DeepONet`	Operator (184 s)	699,528 params; 4-layer branch MLP on $\mathbf{c}$ + 4-layer trunk MLP on $\mathbf{x}$ (width 256, latent 128), MSE loss, AdamW (lr $= 5 \times 10^{-4}$), case-balanced batch 8192.
`PINN data-only`	Coord. (86.8 s)	277,512 params; 5-layer Fourier-coordinate MLP ($B = 4$ bands, width 256, GELU + LayerNorm), MSE loss.
`PINN continuity`	Physics (235 s)	277,512 params; identical to `PINN data-only` + $\lambda_{\text{cont}} = 0.05\,\overline{\mathcal{R}^2_{\text{cont}}}$ on 1024 collocation nodes/step.
`Domain-informed FiLM-MLP`	FiLM (384 s)	1,069,064 params; 5-block residual MLP (width 256, $B = 4$ Fourier bands + leg centerline/arc-length features), per-block FiLM modulation by $\mathbf{c}$, Huber loss ($\delta = 1.0$), AdamW (lr $= 5 \times 10^{-4}$, batch 8192).
`Domain-informed Concat-MLP`	Ablation (111 s)	344,072 params (3.11× smaller than `FiLM-MLP`); identical 5-block width-256 trunk and training, concatenating $\mathbf{c}$ at input without FiLM generators.
`Latent Neural Operator`	Latent (666 s)	583,560 params; cross-attention from Fourier coordinates ($B = 4$) to $M = 24$ latent tokens (width 128, 4 heads, depth 3/2), Huber loss.
`Transolver-style`	Slice (430 s)	715,784 params; 4-block physics-attention network (width 128, 4 heads, $S = 32$ slices), 8192 context / 4096 query nodes/step, AdamW (lr $= 3 \times 10^{-4}$, 2200 steps).
`GAMS-GNN-style`	GNN (746 s)	869,096 params; 4-block message passing (width 96) across $k = 6$ fine kNN graph (8192 nodes) and $(20, 20, 6)$ / $(10, 10, 3)$ voxel graphs, AdamW (lr $= 2 \times 10^{-4}$, 1600 steps).
`F-FNO`	FNO (138 s)	155,888 params; 4 factorized 1D spectral blocks (width 24, 12 modes) on $64 \times 64 \times 16$ voxel grid, AdamW (lr $= 10^{-3}$, 2000 steps).
`U-FNO3D` (`U-FNO`)	U-FNO (125 s)	212,408 params; 3 spectral blocks + 3D U-Net branch (width 24, 12 modes) on $64 \times 64 \times 16$ voxel grid.
`POD-RBF` / `POD-GPR`	ROM (2.4 s)	Within-group train-only SVD (≤ 12 modes, 99.99% energy) + Gaussian RBF (train-only LOO-CV over $\ell \in [0.25, 8], \alpha \in [10^{-6}, 1]$) or GPR ($\ell = 1, \alpha = 10^{-4}$).
`Nearest case`	Lookup (1.8 s)	Copies the closest training case in standardized operating-parameter space within the same group.
`Lattice floor (oracle)`	Voxel ref. (1.1 s)	Specified round-trip voxel reference: splats true test-case CFD values onto the $64 \times 64 \times 16$ grid and trilinearly interpolates back to mesh nodes.

4 Results and Discussion

4.1 Optimism of Node-Random Splits Versus Case-Disjoint Validation

Table 3 and Figure 4 compare 80/20 random node splitting against five-fold case-disjoint cross-validation on the 5,451,341 primary sCO_2 nodes from the paired single-seed (`2026`) split-comparison benchmark (`results/co2_candidate_case_and_random_node_metrics.csv`). Whereas the two low-capacity controls (`Mean baseline` and `Ridge`) show error ratios near 1.0× (0.95×–1.26×) because they cannot memorize local mesh neighborhoods, the three strictly matched-configuration tree models (`Random Forest`, `Extra Trees`, `CatBoost`, which use identical tree counts and depths in both protocols) suffer large error inflation under case-disjoint evaluation on thermodynamic state fields: 6.78 ×–8.25× on density, 11.29 ×–30.08× on gauge pressure, and 5.62 ×–10.24× on temperature in case-macro MAE (3.52 ×–4.05× and 2.95 ×–4.39× in median case nMAE on ρ and T). Including `XGBoost` (which used `max_depth=6` on random-node splits vs. `max_depth=8` on case-disjoint splits) widens the four-tree case-macro MAE inflation range to 6.78 ×–10.37× on ρ and 11.29 ×–87.54× on p_{gauge}. This inflation occurs because random node splits allow high-capacity trees to interpolate case-specific operating levels from adjacent mesh nodes of the same simulation.

Table 3: Error inflation ratios (five-fold case-disjoint error divided by 80/20 random-node error, seed `2026`) separating the three strictly matched-configuration tree models (`Random Forest`, `Extra Trees`, `CatBoost`) from `XGBoost` (depth 8 vs. 6) and the low-capacity `Mean baseline` and `Ridge` controls.

Target field	**3 matched trees (Macro MAE ratio)**	**3 matched trees (Median nMAE ratio)**	`XGBoost` **(MAE / nMAE)**	`Mean` control (MAE / nMAE)	`Ridge` control (MAE / nMAE)
Density ρ	**6.78–8.25**×	**3.52–4.05**×	**10.37**× / **3.34**×	0.98× / 0.90×	1.15× / 1.05×
Gauge pressure p_{gauge}	**11.29–30.08**×	n/a[†]	**87.54**× / n/a[†]	1.03× / n/a	1.26× / n/a
Temperature T	**5.62–10.24**×	**2.95–4.39**×	**7.43**× / **3.17**×	0.95× / 0.72×	1.06× / 1.02×
TED ε	1.02–1.46×	1.01–1.09×	1.29× / 0.91×	0.97× / 0.87×	1.01× / 1.00×
TKE k	1.39–2.99×	1.05–1.22×	2.09× / 0.89×	0.98× / 0.87×	1.10× / 1.00×
Velocity u	1.77–2.76×	1.62–2.03×	2.04× / 1.15×	0.99× / 0.99×	1.04× / 1.05×
Velocity v	1.59–2.23×	1.32–1.74×	1.68× / 0.95×	0.99× / 1.01×	1.03× / 0.99×
Velocity w	1.00–1.02×	1.00–1.02×	0.95× / 0.90×	1.00× / 1.01×	1.00× / 1.02×

[†] Within-case nMAE is suppressed for gauge pressure because $\bar{\sigma}_c = 1.51$ kPa reflects only the within-case vertical pressure variation.

4.2 Three-Seed Case-Disjoint Benchmark Across All Eight CFD Fields

Figure 5 shows the three-seed (`2026`, `2027`, `2028`) five-fold case-macro MAE of all 19 competing models relative to the five-fold `Mean baseline`, and Table 4 reports the detailed five-fold metrics—including the 90th-percentile (P90) case nMAE and the specified `Lattice floor (oracle)` $64 \times 64 \times 16$ voxel round-trip reconstruction reference—for the leading pooled and within-group models on each field (complete 20-row tables for all 19 competing models plus `Lattice floor (oracle)` are in Supplementary Tables S1–S8). Every ratio Macro MAE/Mean-base MAE in Table 4 matches the corresponding cell of Figure 5 to the displayed precision.

Four main findings emerge from Table 4 and Figure 5:

1. **Metric-specific winners on state fields (ρ, T) vs. kinematic/turbulence fields (u, v, w, k, ε):** On density and temperature, case-macro MAE and median case nMAE favor different pooled models because small within-case standard deviations in the small loops ($\sigma_c(T) \approx 0.25$–2.5°C) penalize even modest operating-level bias: `Domain-informed Concat-MLP` achieves the lowest pooled case-macro MAE on both ρ (26.3 kg m^{-3}, 79.7% reduction, P90 nMAE 3.74, $R^2_{\text{pooled}} = 0.936$) and T (5.14°C, 65.5% reduction, ahead of `FiLM-MLP` at 5.34°C and `PINN continuity` at 5.48°C), whereas `DeepONet` achieves the lowest pooled median case nMAE on ρ (0.837 ± 0.160) and `PINN continuity` achieves

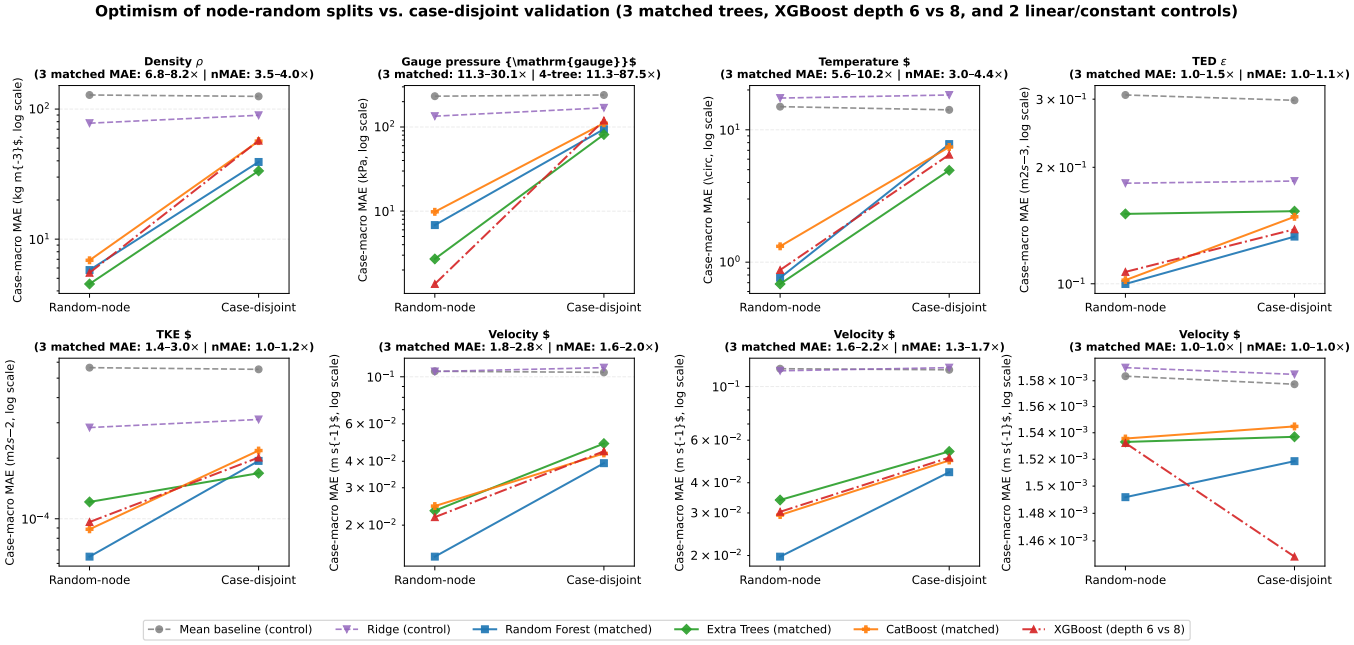


Figure 4: Case-macro MAE (log scale) under 80/20 random-node splits versus five-fold case-disjoint validation across the three strictly matched tree baselines (`RF`, `ET`, `CB`), `XGBoost` (depth 6 vs. 8), and two controls (`Mean`, `Ridge`).

the lowest pooled median case nMAE on T (1.350 ± 0.044; improved to 0.974 ± 0.041 by within-group `POD-RBF`). Across structural groups, `Domain-informed FiLM-MLP` achieves low median nMAE on `VHVC/large` (0.471 on ρ, 0.482 on T, 0.082 on u, 0.088 on v) and `VHVC/small` (0.251 on ρ, 0.500 on T), whereas `HHHC/small` (ρ: 1.391, T: 2.458) and `HHVC/small` (ρ: 2.273, T: 5.363, containing the two near-stagnant 20–30 W cases) drive the P90 tail. On velocity and turbulence fields, `Domain-informed FiLM-MLP` achieves both the lowest case-macro MAE and lowest median case nMAE among all 16 pooled models on u (0.0381 m s^{-1}, nMAE 0.162), v (0.0383 m s^{-1}, nMAE 0.175), w (1.43×10^{-3} m s^{-1}, nMAE 0.272), ε (0.0978 m^2 s^{-3}, nMAE 0.192), and k (0.947×10^{-4} m^2 s^{-2}, nMAE 0.242). When interpolating within a fixed group mesh, `POD-GPR` achieves the lowest overall median case nMAE on ε (0.150 ± 0.011, case-macro MAE 0.0995 m^2 s^{-3}, $R^2_{\text{pooled}} = 0.438$) and `POD-RBF` achieves the lowest median case nMAE on k (0.237 ± 0.015, followed by `FiLM-MLP` at 0.242 and `POD-GPR` at 0.245), although `POD-RBF` exhibits negative R^2_{pooled} on ε (-0.575) and k (-0.070) because unregularized Gaussian RBF extrapolation on boundary cases creates localized turbulence spikes (P90 nMAE 1.46 and 1.25) that inflate node-pooled squared error.

2. **FiLM modulation vs. input concatenation at matched trunk depth/width:** Comparing `Domain-informed FiLM-MLP` (1,069,064 params) against its trunk-matched input-concatenation ablation `Domain-informed Concat-MLP` (344,072 params) shows that adding per-block affine FiLM generators (3.11× total parameter count at identical 5-block, width-256 backbone capacity) reduces median case nMAE on six of the seven spatially varying fields—T (1.467 vs. 1.758), ε (0.192 vs. 0.226), k (0.242 vs. 0.273), u (0.162 vs. 0.224), v (0.175 vs. 0.230), and w (0.272 vs. 0.318)—and lowers case-macro MAE on ε, k, u, v, w, whereas lower-parameter input concatenation yields lower case-macro MAE on ρ (26.3 vs. 31.3 kg m^{-3}) and T (5.14 vs. 5.34°C). Because parameter count is not held constant between the two variants, these gains reflect the combined effect of multiplicative conditioning and the additional FiLM generator parameters.

3. **Gauge-pressure level vs. spatial residual and the zero-residual baseline:** Total gauge-pressure case-macro MAE (83.2 kPa for `Extra Trees`, a 64.7% reduction below the `Mean baseline` of 235.9 kPa, followed by `POD-GPR` at 86.0 kPa and `POD-RBF` at 90.1 kPa) is governed almost entirely

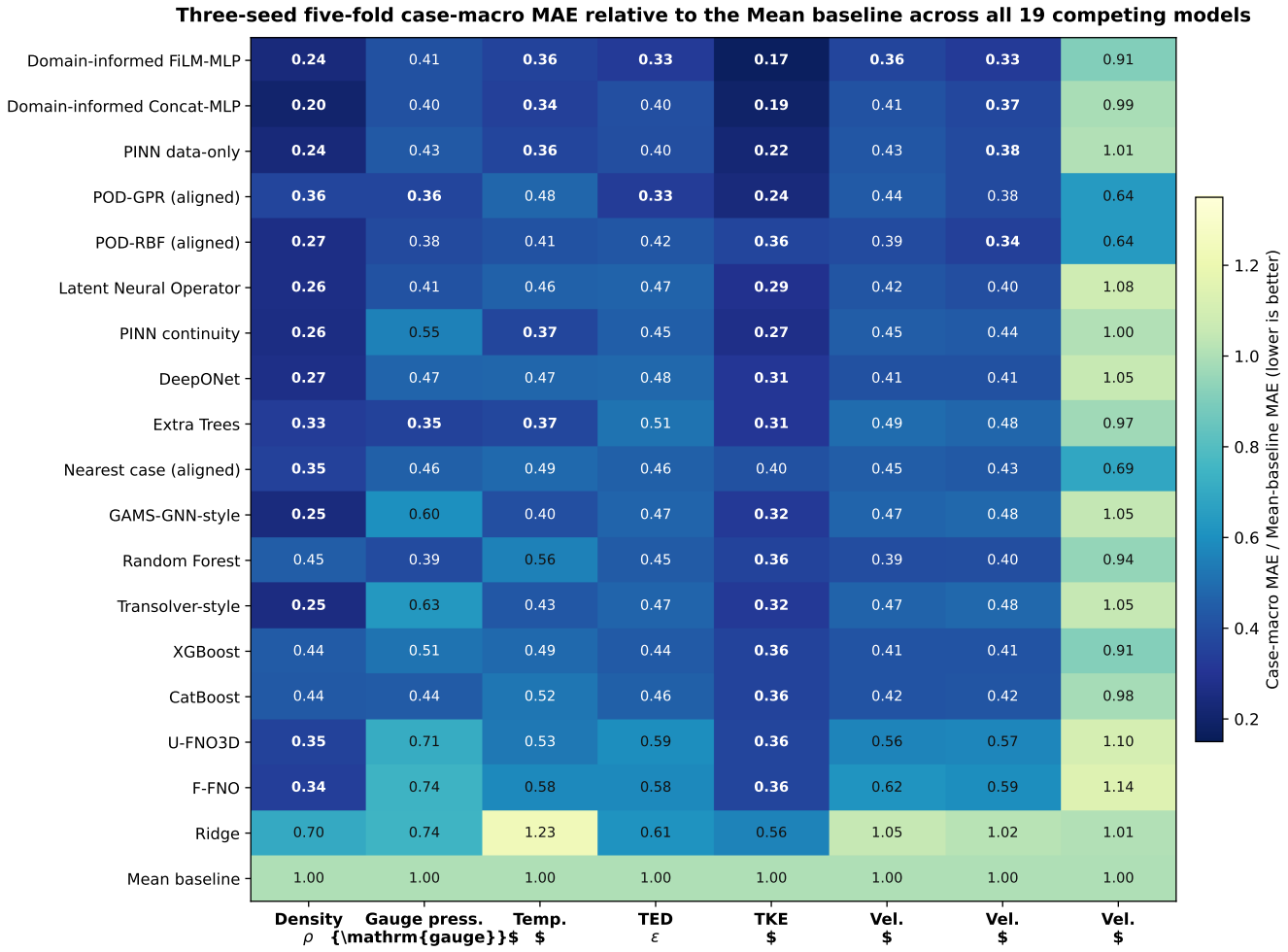


Figure 5: Three-seed mean five-fold case-macro MAE relative to the five-fold `Mean baseline` across all eight CFD fields, ordered by geometric-mean relative error across all 19 competing models (full 20-row tables including the `Lattice floor (oracle)` voxel round-trip reference are in Supplementary Tables S1–S8).

by the case-level offset $e_{\mathrm{level},c}$ (82.9 kPa). On the centered spatial residual $\tilde{p}_{c,i}$ (Table 4 and Supplementary Table S8), the zero-residual baseline ($\hat{\tilde{p}}_{c,i} = 0$, achieved by `Mean baseline`) already attains 42-case RMS $\mathrm{RMSE}_{\mathrm{res,RMS}} = 2.89$ kPa (2886 Pa), 42-case arithmetic-mean $\overline{\mathrm{RMSE}}_{\mathrm{res}} = 1.51$ kPa (1513 Pa), and median $\mathrm{nRMSE}_{\mathrm{res}} = 0.202$. Only four models improve upon the zero-residual baseline across all three residual summaries: within-group `Nearest case` ($\mathrm{RMSE}_{\mathrm{res,RMS}} = 0.67$ kPa, $\overline{\mathrm{RMSE}}_{\mathrm{res}} = 0.30$ kPa, median $\mathrm{nRMSE}_{\mathrm{res}} = 0.0122$), `POD-GPR` (0.46 kPa, 0.22 kPa, 0.0457), `POD-RBF` (0.47 kPa, 0.22 kPa, 0.0483), and pooled `XGBoost` (1.01 kPa, 0.55 kPa, 0.151). Linear `Ridge` achieves a slightly lower $\mathrm{RMSE}_{\mathrm{res,RMS}}$ (2.56 kPa) by partially capturing the vertical hydrostatic residual in `VHVC/large`, but degrades on arithmetic-mean $\overline{\mathrm{RMSE}}_{\mathrm{res}}$ (1.82 kPa) and median $\mathrm{nRMSE}_{\mathrm{res}}$ (1.07). By contrast, coordinate networks and neural operators inject spurious spatial fluctuations of $\mathrm{RMSE}_{\mathrm{res,RMS}} = 8.55$–$60.5$ kPa ($\overline{\mathrm{RMSE}}_{\mathrm{res}} = 7.13$–$37.4$ kPa, median $\mathrm{nRMSE}_{\mathrm{res}} = 4.53$–$24.3$), and multi-output tree ensembles (`Extra Trees`, `Random Forest`, `CatBoost`) yield $\mathrm{RMSE}_{\mathrm{res,RMS}} = 10.1$–$31.9$ kPa ($\overline{\mathrm{RMSE}}_{\mathrm{res}} = 5.49$–$21.9$ kPa, median $\mathrm{nRMSE}_{\mathrm{res}} = 2.91$–$11.1$).

4. **Specified voxel round-trip reconstruction reference and FNO underperformance on slender pipe loops:** Evaluating the specified `Lattice floor (oracle)` round-trip operator (Table 4) shows that splatting ground-truth CFD fields onto the $64 \times 64 \times 16$ Cartesian grid and trilinearly interpolating back to mesh nodes—with zero learning error—already incurs median case nMAE of 0.336

Table 4: Three-seed (2026, 2027, 2028) five-fold out-of-fold accuracy across the 42 sCO_2 cases. For each target, the leading cross-configuration pooled models by case-macro MAE and median case nMAE are listed, followed by the leading within-group mesh-aligned models (POD-RBF, POD-GPR, Nearest case) and the specified Lattice floor (oracle) $64 \times 64 \times 16$ voxel round-trip reference.

Target (unit)	Model	Scope	Macro MAE	Mean-base MAE	Reduction (%)	Macro RMSE	Median nMAE ($\pm$SD)	P90 nMAE	Pooled R^2
ρ (kg m^{-3})	DeepONet	Pooled	35.1	129.3	72.8%	37.1	0.837 ± 0.160	5.60	0.862
	Concat-MLP	Pooled	26.3	129.3	79.7%	29.1	1.011 ± 0.069	3.74	0.936
	Lattice floor	Voxel ref.	4.78	129.3	96.3%	8.15	0.336 ± 0.000	0.461	0.996
p_{gauge} (kPa)	Extra Trees	Pooled	83.2	235.9	64.7%	83.7	2.91†	10.1‡	0.815
	POD-GPR	Aligned	86.0	235.9	63.6%	86.0	0.0457†	0.46‡	0.873
	POD-RBF	Aligned	90.1	235.9	61.8%	90.2	0.0483†	0.47‡	0.571
	Nearest case	Aligned	108.1	235.9	54.2%	108.2	0.0122†	0.67‡	0.776
	Mean baseline	Zero-res	235.9	235.9	0.0%	235.9	0.202†	2.89‡	-0.126
	Lattice floor	Voxel ref.	0.0094	235.9	100.0%	0.0141	0.0018†	0.0268‡	1.000
T (°C)	Concat-MLP	Pooled	5.14	14.90	65.5%	6.89	1.758 ± 0.145	6.57	0.792
	FiLM-MLP	Pooled	5.34	14.90	64.1%	6.79	1.467 ± 0.254	6.40	0.806
	PINN continuity	Pooled	5.48	14.90	63.2%	7.32	1.350 ± 0.044	6.11	0.736
	POD-RBF	Aligned	6.11	14.90	59.0%	7.42	0.974 ± 0.041	4.33	0.620
	Lattice floor	Voxel ref.	0.768	14.90	94.8%	1.24	0.334 ± 0.000	0.452	0.992
ε (m^2 s^{-3})	FiLM-MLP	Pooled	0.0978	0.3002	67.4%	0.315	0.192 ± 0.020	0.474	0.569
	POD-GPR	Aligned	0.0995	0.3002	66.9%	0.330	0.150 ± 0.011	0.905	0.438
	POD-RBF	Aligned	0.127	0.3002	57.8%	0.369	0.177 ± 0.018	1.46	-0.575
	Lattice floor	Voxel ref.	0.171	0.3002	43.2%	0.409	0.339 ± 0.000	0.403	0.072
k (10^{-4} m^2 s^{-2})	FiLM-MLP	Pooled	0.947	5.547	82.9%	1.57	0.242 ± 0.051	0.679	0.898
	POD-GPR	Aligned	1.34	5.547	75.8%	2.12	0.245 ± 0.010	1.27	0.729
	POD-RBF	Aligned	2.00	5.547	64.0%	2.78	0.237 ± 0.015	1.25	-0.070
	Lattice floor	Voxel ref.	2.34	5.547	57.8%	3.74	0.415 ± 0.000	0.446	0.464
u (m s^{-1})	FiLM-MLP	Pooled	0.0381	0.1051	63.7%	0.0569	0.162 ± 0.015	0.810	0.801
	Lattice floor	Voxel ref.	0.0355	0.1051	66.2%	0.0628	0.196 ± 0.000	0.265	0.779
v (m s^{-1})	FiLM-MLP	Pooled	0.0383	0.1175	67.4%	0.0591	0.175 ± 0.004	0.706	0.861
	Lattice floor	Voxel ref.	0.0439	0.1175	62.6%	0.0749	0.241 ± 0.000	0.344	0.793
w (10^{-3} m s^{-1})	FiLM-MLP	Pooled	1.43	1.577	9.3%	3.46	0.272 ± 0.007	0.486	0.418
	POD-GPR	Aligned	1.01	1.577	36.0%	2.56	0.203 ± 0.012	0.400	0.665
	POD-RBF	Aligned	1.01	1.577	35.9%	2.55	0.207 ± 0.008	0.378	0.650
	Nearest case	Aligned	1.08	1.577	31.4%	2.93	0.185 ± 0.009	0.412	0.551
	Lattice floor	Voxel ref.	1.50	1.577	4.9%	4.02	0.323 ± 0.000	0.348	0.159

† For gauge pressure, the Median nMAE column reports the 42-case median centered spatial residual $\text{nRMSE}_{\text{res}}$ (1 kPa denominator floor; Table S8). ‡ For gauge pressure, the P90 column reports the 42-case root-mean-square residual $\text{RMSE}_{\text{res,RMS}}$ in kPa (the corresponding 42-case arithmetic-mean residual $\overline{\text{RMSE}}_{\text{res}}$ in Table S8 is 5.49 kPa for Extra Trees, 0.22 kPa for POD-GPR/POD-RBF, 0.30 kPa for Nearest case, 1.51 kPa for Mean baseline, and 0.0141 kPa for Lattice floor).

(ρ), 0.334 (T), 0.339 (ε), 0.415 (k), 0.196 (u), 0.241 (v), and 0.323 (w). Although learned spectral grid values could in principle partially compensate for trilinear interpolation blur (so Lattice floor (oracle) is a specified round-trip reference rather than a strict mathematical lower bound over all grid-based models), the fact that voxel widths in the large loop (47.0×63.8 mm) are $3.4\times$–$4.6\times$ wider than the pipe diameter ($D_i = 13.84$ mm) causes this round-trip error alone to exceed the out-of-fold

error of `Domain-informed FiLM-MLP` on every velocity and turbulence field, explaining why `F-FNO` and `U-FNO3D` rank near the bottom of the neural models (Fig. 5).

Exploratory statistical comparisons and paired case-level effect sizes: Treating nonparametric Friedman tests and two-sided Wilcoxon signed-rank tests across the 42 cases (computed on three-seed case-averaged nMAE_c for the seven spatially varying fields $\rho, T, \varepsilon, k, u, v, w$ and on case-level MAE in kPa for p_{gauge}, with Holm step-down adjustment across the 18 comparisons against `Domain-informed FiLM-MLP` applied separately within each of the eight fields, accompanied by the median paired difference $\Delta = \mathrm{Median}_c(M_{\mathrm{comp},c} - M_{\mathrm{FiLM},c})$ and its 2,000-resample paired percentile bootstrap 95% confidence interval seeded with `np.random.default_rng(2026)`; Supplementary Table S10) as exploratory, overall model differences are significant on all eight fields ($p < 0.001$). Against `Domain-informed FiLM-MLP` ($\alpha = 0.05$): (i) on ε and k, `FiLM-MLP` ranks first among pooled models (overall mean rank 4.62), significantly outperforming all 15 pooled baselines on ε ($\Delta = +0.034$ $[+0.021, +0.070]$ vs. `Concat-MLP`, $+0.073$ $[+0.056, +0.098]$ vs. `DeepONet`, $+0.124$ $[+0.094, +0.141]$ vs. `Extra Trees`) and 14 of 15 on k ($\Delta = +0.150$ $[+0.077, +0.176]$ vs. `DeepONet`, $+0.186$ $[+0.140, +0.230]$ vs. `Extra Trees`; all except `Concat-MLP`, $\Delta = +0.034$ $[-0.005, +0.059]$, $p_{\mathrm{Holm}} = 0.190$), while not differing significantly from within-group `POD-GPR` ($\Delta = -0.036$ $[-0.069, -0.002]$ on ε, -0.022 $[-0.053, +0.023]$ on k), `POD-RBF` ($\Delta = -0.023$ $[-0.062, +0.013]$ on ε, $+0.012$ $[-0.036, +0.058]$ on k), and `Nearest case` ($p_{\mathrm{Holm}} \geq 0.190$); (ii) on u and v, `FiLM-MLP` ranks first (5.83 and 5.26), significantly outperforming `Concat-MLP` ($\Delta = +0.029$ $[+0.018, +0.055]$ on u, $+0.035$ $[+0.017, +0.066]$ on v), both `PINNs` ($\Delta = +0.066$ and $+0.100$), `LNO`, `Transolver-style`, `GAMS-GNN-style`, both `FNOs` (`U-FNO3D`: $\Delta = +0.153$ $[+0.119, +0.171]$ and $+0.184$ $[+0.147, +0.228]$), `Ridge`, and `Mean baseline` ($p_{\mathrm{Holm}} < 0.05$, plus `DeepONet` $\Delta = +0.086$ and `Extra Trees` $\Delta = +0.103$ on v), while not differing significantly from the remaining tree ensembles or within-group aligned models ($p_{\mathrm{Holm}} \geq 0.052$); (iii) on w, `FiLM-MLP` is significantly outperformed by within-group `POD-RBF` ($\Delta = -0.097$ $[-0.136, -0.069]$), `POD-GPR` ($\Delta = -0.087$ $[-0.124, -0.069]$), and `Nearest case` ($\Delta = -0.076$ $[-0.094, -0.064]$), does not differ significantly from `PINN continuity`, tree ensembles, `Ridge`, or `Mean baseline` ($p_{\mathrm{Holm}} \geq 0.096$), and significantly outperforms all other neural/operator baselines ($\Delta = +0.041$ $[+0.024, +0.051]$ vs. `Concat-MLP`, $p_{\mathrm{Holm}} < 0.05$); and (iv) on ρ, T, and p_{gauge}, `FiLM-MLP` is significantly outperformed by `POD-RBF` on T ($\Delta = -0.708$ $[-1.382, -0.048]$) and by `Extra Trees` ($\Delta = -121.4$ kPa $[-221.3, -35.6]$), `POD-GPR` ($\Delta = -118.6$ kPa), `POD-RBF` ($\Delta = -114.4$ kPa), and `Random Forest` ($\Delta = -110.2$ kPa) on p_{gauge} case MAE ($p_{\mathrm{Holm}} \leq 0.024$), while not differing significantly from `DeepONet` ($\Delta = +0.076$ on ρ, $+0.151$ on T) or `Concat-MLP` ($\Delta = -0.060$ on ρ, $+0.094$ on T; $p_{\mathrm{Holm}} \geq 0.170$).

4.3 Spatial Reconstruction Analysis and Leave-One-Group-Out Shift

Figure 6 shows held-out CFD fields, surrogate predictions, and absolute error maps (seed `2026`, labeled by nMAE_c rank) for showcase cases on ρ (`DeepONet`, rank 15/42; 3-seed rank 3/42), T (`PINN continuity`, rank 17/42; 3-seed rank 1/42), and v (`FiLM-MLP`, rank 10/42; 3-seed rank 3/42), while Figure 7 directly presents the corresponding **median (rank 21/42)** and **worst (rank 42/42)** held-out cases per field in the main text. The median (rank 21/42) cases capture loop-scale buoyancy stratification and vertical core velocity gradients with errors concentrated along the cooling-jacket boundary layer and pipe elbows, whereas the worst (rank 42/42) cases occur at the highest-power, hottest boundary condition in the dataset (`HHHC_P2000_90_bar_0.05kgs_T_315` on ρ and T: $Q = 2000$ W, $p_{\mathrm{op}} = 90$ bar, $\dot{m} = 0.05$ kg s^{-1}, $T_{\mathrm{in}} = 315$ K, one of three $T_{\mathrm{in}} = 315$ K `HHHC` cases and the hottest case overall at $\bar{T}_c = 47.5$°C) or in the near-stagnant `HHVC_P20_90_bar_0.05kgs_T_305` run on v ($\sigma_c(v) = 1.5$ mm s^{-1}).

Table 5 summarizes the leave-one-group-out shift study across the four structural/regime groups (16 pooled models × 8 fields = 128 model–field evaluations per group; full 16-model metrics in Supplementary Table S9). To avoid post-hoc selection bias from reporting only the retrospective winner per cell, Table 5 reports a **fixed seven-model display set** (`Mean baseline`, `Extra Trees`, `XGBoost`, `DeepONet`, `PINN continuity`, `Domain-informed Concat-MLP`, and `Domain-informed FiLM-MLP`) alongside the best observed pooled model across all 16 candidates. When `VHVC/small` ($n = 4$) is held out, both the small-loop geometry (0.894 × 1.000 m, 90 bar, 305 K) and the `VHVC` orientation are represented in the training set,

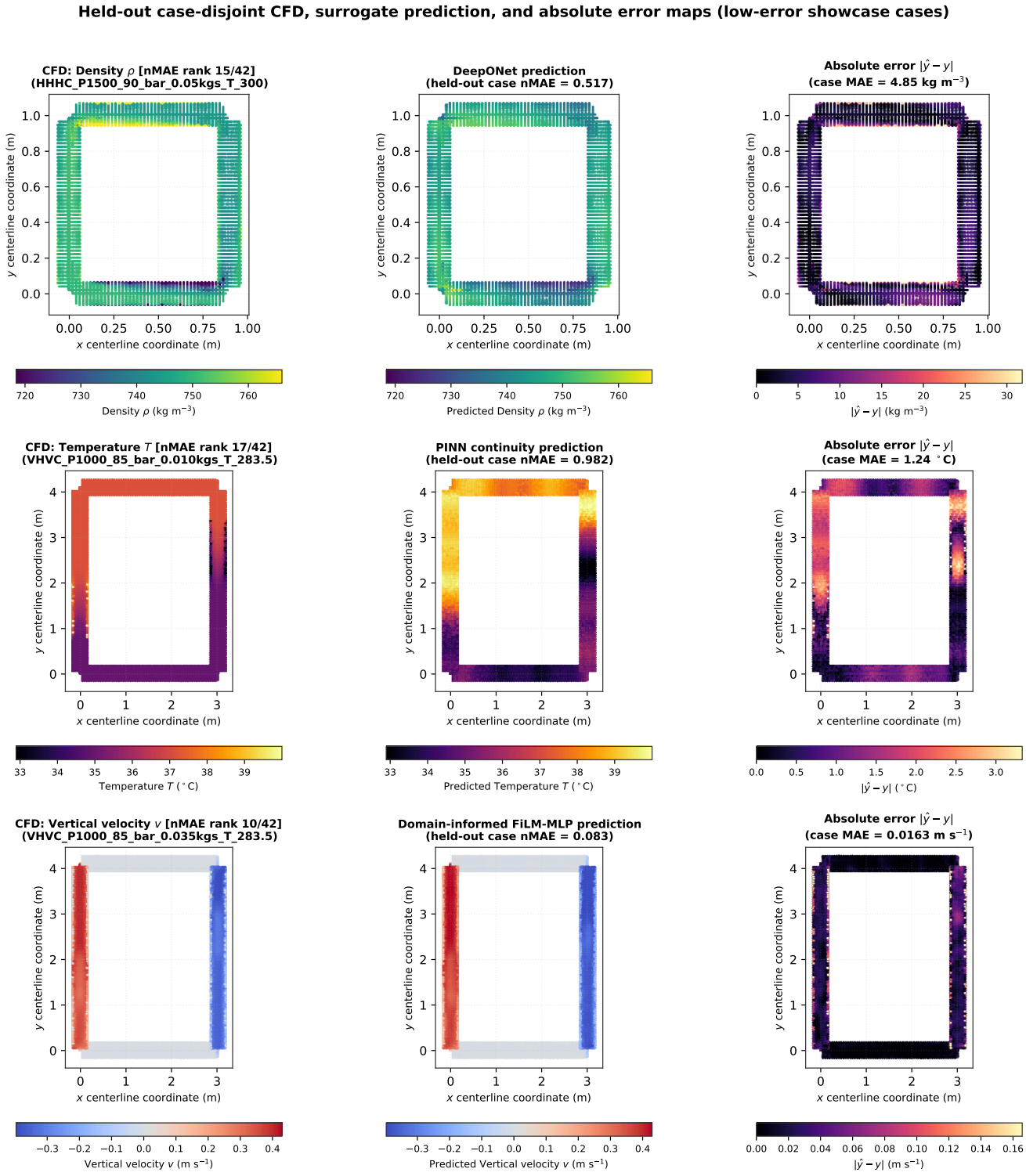


Figure 6: Showcase held-out out-of-fold spatial reconstructions (seed 2026) comparing reference CFD (left), surrogate prediction (middle), and absolute error $|\hat{y} - y|$ (right) for ρ (`DeepONet`, rank 15/42), T (`PINN continuity`, rank 17/42), and v (`Domain-informed FiLM-MLP`, rank 10/42).

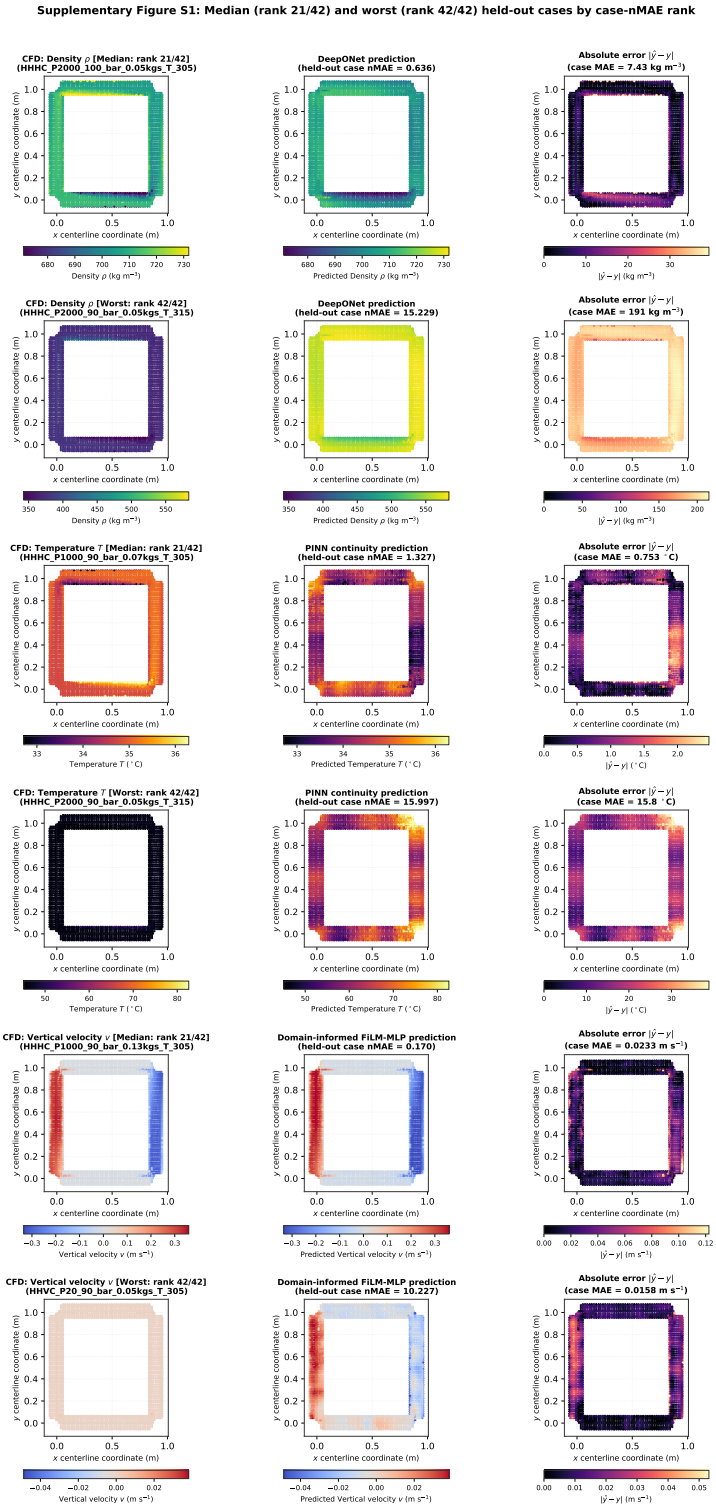


Figure 7: Median (rank 21/42, left block) and worst (rank 42/42, right block) held-out out-of-fold spatial reconstructions (seed `2026`) for ρ (`DeepONet`), T (`PINN continuity`), and v (`Domain-informed FiLM-MLP`). Rank-42/42 failures correspond to the highest-power $T_{\text{in}} = 315$ K case (`HHHC_P2000_90_bar_0.05kgs_T_315`, $Q = 2000$ W, 90 bar, 0.05 kg s^{-1}, $\bar{T}_c = 47.5$°C, one of three 315 K `HHHC` runs) on ρ, T and the near-stagnant 20 W `HHVC` case on v.

enabling XGBoost, Extra Trees, and DeepONet to improve substantially upon the Mean baseline across ρ, T, ε, k, and v, whereas Domain-informed FiLM-MLP improves on k (1.09 vs. 6.79) but degrades relative to Mean baseline on ρ (7.42 vs. 4.52), T (16.4 vs. 8.51), ε (14.4 vs. 9.04), and v (1.22 vs. 1.08). Conversely, holding out VHVC/large ($n = 13$, shifting loop dimensions by 3.4 × –4.1× and T_{in} to 283.5 K) or HHHC/small ($n = 20$, removing all horizontal-cooler cases at Q = 1000–2000 W) leaves most models close to or worse than Mean baseline, with only modest field-specific gains on VHVC/large by Random Forest (ρ: 3.85 vs. 5.57; ε: 0.42 vs. 0.43), Extra Trees (ρ: 4.36, T: 1.65 vs. 1.70, v: 0.77 vs. 0.84), and PINN data-only (k: 1.16 vs. 1.89). On VHVC/large, training exclusively on the 29 small-loop cases leaves the geometric descriptors (L_x, L_y, D_i) with zero variance in the training split (triggering the cf_scale = 1.0 fallback), so evaluating on the 3.4×–4.1× larger loop drives unmodulated linear and spectral models (Ridge, Concat-MLP, F-FNO, U-FNO3D, and DeepONet on ε, k) into the np.clip(z, -12.0, 12.0) safety clamp of TargetScaler.inverse on arcsinh-scaled turbulence fields ($\sinh(12) \approx 8.14 \times 10^4$, producing identical saturated errors near 1.62×10^9 on ε in Supplementary Table S9; denoted $>10^3$ in Table 5), whereas FiLM-MLP avoids clamp saturation (ρ: 5.29 vs. 5.57, k: 1.87 vs. 1.89) but still degrades substantially on T (2.50 vs. 1.70) and ε (25.0 vs. 0.43).

Table 5: Leave-one-group-out structural/regime shift benchmark across the four groups (16 pooled models × 8 fields = 128 evaluations per group). Top rows report the three-seed mean median case nMAE on $(\rho, T, \varepsilon, k, v)$ for a fixed seven-model display set; the bottom block lists the retrospective best pooled model per field (p_{gauge} via level MAE in kPa and $\text{nRMSE}_{\text{res}}$).

Block / Model	Field(s)	HHHC/small ($n = 20$)	HHVC/small ($n = 5$)	VHVC/small ($n = 4$)	VHVC/large ($n = 13$)
Fixed seven-model display set: median case nMAE tuple (ρ / T / ε / k / v)					
Mean baseline	$\rho/T/\varepsilon/k/v$	5.11/9.57/5.85/4.63/0.81	13.3/21.8/15.7/9.95/0.95	4.52/8.51/9.04/6.79/1.08	5.57/1.70/0.43/1.89/0.84
Extra Trees	$\rho/T/\varepsilon/k/v$	7.00/11.6/0.41/1.03/1.01	1.17/3.34/0.33/0.62/1.21	2.25/3.74/0.37/0.99/0.89	4.36/1.65/0.44/1.98/0.77
XGBoost	$\rho/T/\varepsilon/k/v$	11.9/18.1/7.51/10.7/1.19	1.39/3.05/0.31/0.52/1.41	**0.98**/**1.61**/0.28/0.47/**0.69**	5.85/1.73/7.22/1.83/0.79
DeepONet	$\rho/T/\varepsilon/k/v$	9.66/68.8/0.41/0.97/0.99	6.11/7.94/0.39/0.70/1.49	3.50/7.18/0.37/0.85/0.78	5.90/2.51/ $>10^3$/ $>10^3$/0.82
PINN continuity	$\rho/T/\varepsilon/k/v$	5.48/44.2/1.01/1.58/0.73	4.94/19.2/0.49/0.66/1.11	6.57/12.5/18.9/2.13/0.77	5.36/3.18/4.46/2.47/0.91
Concat-MLP	$\rho/T/\varepsilon/k/v$	12.9/130/1.29/1.31/0.88	17.1/34.7/29.5/1.60/1.53	3.24/11.6/ $>10^3$/14.0/2.53	$>10^3$/ $>10^3$/ $>10^3$/ $>10^3$/ $>10^3$
FiLM-MLP	$\rho/T/\varepsilon/k/v$	12.1/113/6.82/1.30/1.05	12.1/13.9/0.90/0.88/1.14	7.42/16.4/14.4/1.09/1.22	5.29/2.50/25.0/1.87/0.86
Retrospective best pooled model across all 16 candidates (Supplementary Table S9)					
Best on ρ / T	nMAE	GNN (4.60) / Mean (9.57)	ET (1.17) / XGB (3.05)	XGB (0.98) / XGB (1.61)	RF (3.85) / ET (1.65)
Best on ε / k	nMAE	F-FNO (0.40) / F-FNO (0.65)	RF (0.28) / Ridge (0.44)	XGB (0.28) / XGB (0.47)	RF (0.42) / PINN-d (1.16)
Best on u / v / w	nMAE	PINN-c (0.69) / PINN-d (0.72) / Mean (0.38)	CB (0.79) / CB (0.67) / Mean (0.38)	XGB (1.12) / XGB (0.69) / Mean (0.39)	XGB (0.59) / ET (0.77) / Mean (0.23)
Best on p_{gauge}	Level / $\text{nRMSE}_{\text{res}}$	GNN (123) / Mean (0.196)	XGB (557) / Mean (0.186)	Mean (128) / Mean (0.185)	LNO (248) / XGB (0.988)

4.4 Limitations

Three limitations of this study should be noted: (1) because the dataset is curated from legacy converged CFD-Post exports without original solver case/journal files or convergence histories, we cannot separate a physical flow-regime transition from convergence to an alternative steady numerical branch in the two near-stagnant cases HHVC_P20_90_bar_0.05kgs_T_305 (Q = 20 W) and HHVC_P30_90_bar_0.05kgs_T_305 (Q = 30 W), nor can we directly verify the specific k–ε sub-variant, wall-function treatment, or inferred hydrostatic reference density ρ_0; (2) plane-weighted mean CO_2 density $\bar{\rho}_{\text{plane},c}$ (averaged over exported sampling planes rather than volume-integrated over the pipe) varies across steady-state cases at the same $(p_{\text{op}}, T_{\text{in}})$ (Section 2.2), so the cases represent independent steady operating points rather than a verified constant-mass inventory trajectory; and (3) seeds 2026, 2027, and 2028 re-randomize the five-fold stratified case splits in addition to model initialization, and the small group sizes in HHHC/small ($n = 5$) and VHVC/small ($n = 4$) limit statistical power under group holdout.

5 Conclusions

We have presented a case-disjoint surrogate benchmark across 42 steady-state sCO_2 natural-circulation loop CFD simulations (5,451,341 primary sCO_2 nodes). Our main conclusions are:

1. **Case-disjoint validation and two-part pressure evaluation are essential:** Across the three strictly matched-configuration tree baselines (RF, ET, CB), 80/20 random node splits understate five-

fold case-disjoint case-macro MAE by 6.78 × –8.25× on density, 11.29 × –30.08× on gauge pressure, and 5.62 × –10.24× on temperature (6.78 × –10.37× and 11.29 × –87.54× across all four tree baselines including `XGBoost`), while `Mean` and `Ridge` controls remain near 0.95 × –1.26×. Separating gauge pressure into its case-level mean offset and centered spatial residual—referenced against the zero-residual baseline ($\mathrm{RMSE}_{\mathrm{res,RMS}} = 2.89$ kPa, $\overline{\mathrm{RMSE}}_{\mathrm{res}} = 1.51$ kPa, median $\mathrm{nRMSE}_{\mathrm{res}} = 0.202$)—shows that only `Nearest case`, `POD-GPR`, `POD-RBF`, and `XGBoost` improve upon predicting zero spatial residual across all three residual summaries.

2. **Different physical fields and metrics favor distinct inductive biases:** Under three-seed five-fold case-disjoint cross-validation, `Domain-informed FiLM-MLP` leads all pooled models in both case-macro MAE and median case nMAE across all three velocity components (nMAE = $0.162, 0.175, 0.272$) and both turbulence fields (nMAE = 0.192 on ε, 0.242 on k); on density and temperature, `Domain-informed Concat-MLP` achieves the lowest pooled case-macro MAE (26.3 kg m^{-3} and 5.14°C) whereas `DeepONet` (nMAE = 0.837) and `PINN continuity` (nMAE = 1.350) achieve the lowest pooled median case nMAE; and within-group mesh-aligned `POD-RBF` and `POD-GPR` achieve the lowest median case nMAE on temperature (0.974), ε (0.150 for `POD-GPR`; 0.177 for `POD-RBF`), and k (0.237 for `POD-RBF`; 0.245 for `POD-GPR`) when interpolating on a fixed mesh, whereas uniform voxelized spectral operators (`F-FNO`, `U-FNO`) are severely limited by coarse 64 × 64 × 16 Cartesian voxelization (`Lattice floor (oracle)` round-trip nMAE = 0.196–0.415) on slender pipe geometries ($D_i/L < 1.7\%$).

3. **Cross-group generalization requires multi-scale coverage:** Leave-one-group-out tests (128 model–field evaluations per group) show that pooled tree and neural operators (`XGBoost`, `Extra Trees`, `DeepONet`) beat the `Mean baseline` across $\rho, T, \varepsilon, k, v$ when the loop scale and orientation are covered in training (`VHVC/small`), whereas under combined 3.4 × –4.1× geometric and operating-regime shifts (`VHVC/large`, `HHHC/small`) most surrogates degrade close to or worse than the `Mean baseline`.

Data and Code Availability

All preprocessing, benchmark training, statistical testing, and plotting scripts, along with the three-seed metric tables, pressure/density audit CSV, and supplementary material (Section S1, Figure S1, and Tables S0a–S10), are publicly available at https://github.com/Barghav777/SCWR_3D.

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